

Vineet Singh

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EXPERIENCE

Samsung Research Institute - Delhi

Lead Engineer — Tizen (Contractor via Tech Mahindra)

Noida, UP

Oct 2025 – Present

- Built an **internal Retrieval-Augmented Generation (RAG) assistant** using **1M+ records** spanning content-partner correspondence, error logs, build submissions, and test cases to retrieve historical guidance on Tizen app development and submission issues.
- Developed **AI-powered full-stack tools** using **Python backends, React, and local Hugging Face models** for semantic analysis, clustering, summarization, and translation; integrated a content-recommendation pipeline with a **Samsung browser backend** for internal use.
- Developing an **on-device AI proof of concept** for Samsung TVs with **1 GB RAM**, evaluating lightweight models and exploring **model distillation** for memory-constrained inference.
- Own **Samsung-side integration and build delivery** for **Apple TV, Spotify, and Netflix** on global hospitality TVs, **JioHotstar and ZEE5** on India hospitality TVs, and **iPrep** on education TVs; develop **C++ native integration code** with content partners to adapt retail-TV apps to hospitality/education Tizen environments.
- Extended **Bixby voice-interaction handling** for **ontitleselection**, enabling Tizen TV applications to process spoken title-selection commands through Samsung's VoiceInteraction API.
- Debug partner application issues on hospitality and education TVs across **Tizen web/native boundaries**, using logs and tracing to investigate integration, playback, rendering, and memory-related failures.

LTIMindtree (now LTM)

Senior Software Engineer

Noida, UP

Apr 2023 – Oct 2025

- Built high-throughput backend services using **Java, Spring Boot, Spring Cloud**
- Designed Redis caching layer reducing latency by **45%** under production load
- Implemented reliable MongoDB persistence with replication and indexing strategies
- Developed REST services and async worker handling concurrent financial workloads

Software Engineer — Pune, MH

June 2022 – April 2023

- Built small Node.js utilities** for logging and event-driven monitoring.
- Collaborated on React.js UI enhancements** and bug fixes, improving usability and design consistency.
- Containerized services using **Docker** and contributed to **Kubernetes-based deployment pipeline**.
- Practiced Agile methodologies through sprints, retrospectives, and continuous feedback cycles.

BitWise Inc.

Software Engineer Intern

Pune, MH

Sept 2021 – Jun 2022

- Implemented REST APIs and optimized database queries improving latency

PROJECTS

AI News Aggregator

React, JavaScript, Python, FastAPI, HuggingFace, Vector Embeddings

2025 – Present

- Designed a configurable web crawling pipeline parsing domain rules from **site.json** to extract time-bounded keyword-matched articles across multiple news sources.
- Built local ML inference pipeline using HuggingFace models: BART for abstractive summarization and MiniLM embeddings for semantic representation.
- Implemented vector-similarity clustering to deduplicate cross-source reports of the same event and group semantically equivalent articles into unified UI entities.
- Developed full-stack system (React + FastAPI) with asynchronous processing, reducing redundant content and enabling fast searchable summarized news export.
- Source:** [GitHub](#) [Live Demo](#)

TECHNICAL SKILLS

Systems: C++, Tizen OS, Linux, Native Integration, Runtime Debugging, Memory Analysis

Backend: Python, FastAPI, Java, Spring Boot, Spring Cloud, Distributed Systems, REST APIs, Node.js

Web & Voice: JavaScript, React, Bixby Integration, Samsung VoiceInteraction API

AI/ML: RAG, Hugging Face, Vector Embeddings, NLP, Semantic Clustering, Local Model Inference, On-Device Model Evaluation

Data & Infra: MongoDB, PostgreSQL, Redis, Docker, Kubernetes, AWS, GCP

CODING

LeetCode: 1820 — **CodeChef:** 1892 — [Portfolio](#)

EDUCATION

GL Bajaj Institute of Technology and Management

Bachelor's of Technology in Computer Science and Engineering

Noida, UP

Aug. 2018 – July 2022